From a verified single-coin obstruction to the population problem

Research checkpoint: proof, counterexample, and scope audit

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Abstract

We verify a sharp critical-margin obstruction for single-coin estimators and investigate its population implications. For the exact quadratic magnitude law on $(0, 1/4]$, separate uniform variance and expected-work suprema have product of order $\rho L(1 + \log L)$, where $L = 1 + \log(1/\rho)$, even with uniform bias at most $\rho/64$. This result was inherited from concurrent repository work; the current checkpoint independently audits it and refines one proof. It excludes budget-dependent raw iid averages but does not exclude nonlinear population estimators. A finite-prefix directional-information reduction handles arbitrary creation and interleaving of coins. An explicit observable gambler's-ruin procedure disproves the extra-log information upper bound on the natural radial submodel; calibration attains the rate without that logarithm on the specified submodel. A separate fixed-feature theorem excludes improvement from independent pilot selection of a scalar control variate. Exact rational finite-horizon diagnostics exhibit small benefits from adaptive allocation. The arbitrary-law rare-event minimax problem remains unresolved.

Evidence discipline. The statistical assertions here are **Proved-on-paper** with separate adversarial agent review, not Lean formalized or externally peer reviewed. The finite computations are **Numerical**, using exact rational arithmetic in their stated finite scope. The radial information upper bound is **Disproved** by an analytic counterexample. The full minimax formula is a **Conjecture**. Priority is unresolved; no novelty claim is made. The independent replay of nine exact Tier5 declarations applies only to their supplied realized prefix laws and balanced sign priors, not to these rare-event theorems.

1 Experiment and question

For each fresh iid coin, $D \sim \mu$ on $(0, 1/2]$ is hidden and $S \mid D = d \sim \mathrm{Bernoulli}(q(d))$. Conditional flips are iid with success probability $1/2 + (2S - 1)D$. The unknown measurable allocation belongs to the global class $\mathcal{Q}_\rho = \{0 \le q \le 1 : \int q\, d\mu \le \rho\}$; the target is $\theta(q) = \int q\, d\mu$. Knowing μ, ρ does not reveal any coin's D, S. Algorithms observe bits and coin identities, may create, revisit and interleave coins, and may stop adaptively. Every genuine flip costs one, including pilot flips. Deterministic arithmetic and independent randomness are free. Exact deterministic integrals of the specified known law are allowed in theoretical constructions; approximate implementations need a separate numerical error analysis. Uniform expected work is $\sup_q \mathbb{E}_q T \le B$, not a deterministic cap.

The candidate arbitrary-law functional is

$$\Psi_{\mu,\rho}(h) = \sup_{0 \le w \le 1,\ \int w\, d\mu \le \rho} \int \min\{w(d), h\sqrt{w(d)}/d\}\, \mu(dd).$$

For the exact quadratic law with $a = 1/4$, its unsaturated order is $\sqrt{\rho L/B}$ at $h = B^{-1/2}$; the currently established upper construction for arbitrary q retains the factor $\sqrt{1 + \log L}$. The single-coin obstruction alone does not decide which rate is the full population minimax rate. Confidence-radius optimality and uniform adaptation to unknown μ, ρ are not established in this checkpoint.

2 What was independently verified

The inherited full proof appears in the final appendix, including its finite-gap entropy lemma, arbitrary thinning argument, global-rarity localization, small-bias extension, and matching single-coin upper bound. Its source snapshot has SHA-256 beginning `29463b56bfcb`; the earlier version first inspected by the reviewer began `d85f58c1201f`. The added small-bias section was reviewed separately. The original sources and v4 paper were not edited. Chen--Li--Qiao's scale-entropy mechanism and interior Bernoulli applicability are prior art; the focused source audit does not establish priority for the product theorem.

The next section records a refined small-bias proof and the exact arithmetic-average consequence. The later population reduction identifies the hypothesis actually needed to transfer a one-coin information bound, and the radial construction shows that a tempting version of that hypothesis is false.

3 The single-coin obstruction survives uniformly small bias

PROVED-ON-PAPER; INDEPENDENT VERIFICATION/REFINEMENT. The small-bias extension was already present in the concurrently maintained critical-two research notes when the separate reviewer first inspected them. This independent derivation is not counted as discovery of a new theorem. It is a single-coin obstruction, not a theorem about unrestricted population algorithms. The finite-gap sign entropy lemma below is the sole imported mathematical input. Its separate audit is part of the companion review. This section is not Lean formalized or externally peer reviewed.

Fix $a = 1/4$, $\mu(dd) = 2d/a^2\,dd$ on $(0, a]$, $0 < \rho \le 1/4$, $L = 1 + \log(1/\rho)$ and $H = 1 + \log L$. One genuine coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Bernoulli}(q(d))$; its observations are conditionally iid $\mathrm{Bernoulli}(1/2 + (2S-1)D)$. Let $\mathcal{Q}_\rho = \{q : 0 \le q \le 1, \int q\,d\mu \le \rho\}$ and $\theta(q) = \int q\,d\mu$. An observable single-coin procedure may use an independent seed, output any real Y, and stop at $T = 0$ or at an unbounded stopping time. Only its genuine coin flips are charged. The seed and policy cannot depend on q or the realized hidden gap.

Theorem 3.1. *There is a universal $c > 0$ such that every such procedure with*

$$V := \sup_{q\in\mathcal{Q}_\rho} \mathrm{Var}_q(Y) < \infty, \qquad W := \sup_{q\in\mathcal{Q}_\rho} \mathbb{E}_q T < \infty, \qquad \sup_{q\in\mathcal{Q}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \epsilon \le \rho/64$$

satisfies $VW \ge c\rho LH$. The same lower bound holds for MW, where $M := \sup_q \mathbb{E}_q Y^2$.

The finite-gap input. The following statement is used, with the common randomized test independent of the unknown gap. If $0 < d_K < \cdots < d_1 \le 1/128$, $d_j/d_{j+1} \ge 2$, $K \ge \lceil e^{32} \rceil$, and one observable sign test has error at most $1/10$ at every $1/2 \pm d_j$ and finite expected stopping time at every one of these $2K$ laws, then

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j}\tau \ge \frac{1}{512} K \log K. \tag{N1}$$

No stopping or integrability is required at the fair-coin law. The companion single-coin proof establishes this from finite-prefix KL, time-window overlap, and the binary data-processing inequality. In particular this is not a direct appeal to a pointwise or limsup version of a sign-testing bound.

Proof of Theorem 3.1. First $M \le V + (\rho + \epsilon)^2 < \infty$. Write $m_s(d) = \mathbb{E}_{d,s} Y$ and $M_s(d) = \mathbb{E}_{d,s} Y^2$ on a common full-measure set. Such a set exists because the laws $q = 0$ and $q = \rho$ make both conditional second moments integrable. Set $g = m_+ - m_-$ and $b_q = \mathbb{E}_q Y$. For every feasible q,

$$\left| \int q(g-1)\,d\mu \right| = |b_q - b_0 - \theta(q)| \le 2\epsilon. \tag{N2}$$

Taking $q = \rho$ gives $\int g\,d\mu \ge 31/32$. Conditional variance under this constant sign allocation and Cauchy--Schwarz imply

$$V \ge \rho(1-\rho) \int g^2\,d\mu \ge \frac{2883}{4096}\rho > \frac{7}{10}\rho. \tag{N3}$$

Indeed conditioning on D and then the two signs contributes $\rho(1-\rho)g(D)^2$ to the variance. Since $(\rho+\epsilon)^2 \le (4225/16384)\rho$, it follows that

$$M \le \tfrac{3}{2}V. \tag{N4}$$

We next remove artificial thinning before the first observation. The event $A = \{T \ge 1\}$ is determined by the seed alone, so its probability p is the same for every q and every fixed signed gap. If $p = 0$, the expectation would be constant in q, contradicting the bias guarantee at $\theta = 0$ and $\theta = \rho$; hence $p > 0$. Let $c_0 = \mathbb{E}[Y\mathbf{1}_{A^c}]$, also independent of q. Condition the seed on A, run the original procedure, and output $Z = pY + c_0$. Resampling seeds until A occurs implements this conditioning almost surely using free independent randomness; the constants p, c_0 are fixed by the original policy and may be granted in this lower-bound argument. The new procedure always takes at least one genuine flip, preserves each expectation b_q, and has $W_Z = W/p \ge 1$. For $p < 1$, Cauchy--Schwarz on A^c gives

$$\begin{aligned}\mathbb{E}_q Z^2 &= p\mathbb{E}_q[Y^2\mathbf{1}_A] + 2c_0 b_q - c_0^2 \\ &\le pM + 2c_0 b_q - \frac{c_0^2}{1-p} \le pM + (1-p)b_q^2 \le pM + (\rho+\epsilon)^2.\end{aligned}$$

For $p = 1$ the last bound is immediate. Apply (N3) also to Z, whose bias bound is unchanged. The last display implies $pM \ge (7/10 - 4225/16384)\rho \ge 2\rho/5$. Set

$$P = MW, \qquad m = \max\{\rho, pM + (\rho+\epsilon)^2\}.$$

The preceding inequalities imply

$$\rho \le m \le \tfrac{5}{2}pM \le \tfrac{5}{2}P, \qquad mW_Z \le \tfrac{5}{2}P, \qquad P \ge \tfrac{2}{5}\rho. \tag{N5}$$

In what follows m_s, M_s, w_s denote the conditional means, second moments, and expected work of the active procedure Z. They are finite for both signs on a common full-measure set. Its uniform second moment is at most m, and (N2) remains valid.

Good gaps without exact unbiasedness. For $E = \{g < 7/8\}$, nonatomicity allows a subset of mass exactly ρ if $\mu(E) \ge \rho$. Its indicator would contradict (N2), since $\rho/8 > 2\epsilon$. Thus $\mu(E) < \rho$; applying (N2) to $q = \mathbf{1}_E$ now gives

$$\mu(E) \le 16\epsilon \le \rho/4.$$

The other bad sets are $E_- := \{M_- > 1/64\}$ and $E_+ := \{M_+ - M_- > 2m/\rho\}$. Markov's inequality under $q = 0$ gives $\mu(E_-) \le 64m$. If $\mu(E_+) \ge \rho$, a subset F of mass ρ and $q = \mathbf{1}_F$ would give $\mathbb{E}_q Z^2 = \int M_-\, d\mu + \int_F (M_+ - M_-)\, d\mu > 2m$, contradicting the uniform second-moment bound. Hence $\mu(E_+) < \rho$. On the complement G of these sets and the common null set,

$$\mu(G^c) \le 66m, \quad |m_-| \le 1/8, \quad m_+ \ge 3/4, \quad M_- \le 1/64, \quad M_+ \le 1/64 + 2m/\rho. \tag{N6}$$

Run the active procedure $n = \lceil 256(1+m/\rho)\rceil \le 513m/\rho$ times, with independent seeds on successive unused blocks of the same fixed-gap coin. Compare the average output to $7/16$. At every gap in G the distance of either mean from this threshold is at least $5/16$. Conditional independence of successive blocks follows by summing over the finite stopping prefixes; their conditional finite expected lengths imply almost-sure termination. Chebyshev's inequality gives, for either sign, an error at most

$$\frac{256}{25n}\left(\frac{1}{64} + \frac{2m}{\rho}\right) \le \frac{2}{25} < \frac{1}{10}.$$

The negative expected work of this common test is $nw_-(d)$.

Select separated gaps and apply the entropy bound. For $j \geq 3$, put $I_j = (a4^{-j}/2, a4^{-j}]$, so $\mu(I_j) = (3/4)16^{-j}$. Set

$$J = \left\lfloor \log_{16} \frac{1}{176m} \right\rfloor, \qquad K = J - 2.$$

When $K \geq \lceil e^{32} \rceil$, every $j = 3, \ldots, J$ satisfies $\mu(I_j) \geq 132m$, so at least half its mass is in G. Choose $d_j \in G \cap I_j$ with

$$w_-(d_j) \leq \frac{4}{\mu(I_j)} \int_{I_j} w_- \, d\mu.$$

Such points exist by averaging; conditional finite moments and work may also be imposed because the discarded set is null. These gaps have ratios strictly greater than two and maximum at most $1/256$. Moreover

$$\sum_{j=3}^{J} d_j^2 w_-(d_j) \leq \frac{16a^2}{3} \sum_{j=3}^{J} \int_{I_j} w_- \, d\mu \leq \frac{16a^2}{3} W_Z.$$

Applying (N1) and (N5),

$$\frac{1}{512} K \log K \leq \frac{16a^2}{3} n W_Z \leq 6840 a^2 \frac{P}{\rho}. \tag{N7}$$

It remains to compare scales without assuming the variance is small. If $P \geq \sqrt{\rho}$, then $P \geq c_1 \rho LH$ for a universal $c_1 > 0$, since $\sqrt{\rho} LH$ is bounded on $(0, 1/4]$. If $P < \sqrt{\rho}$, then $m < (5/2)\sqrt{\rho}$, and therefore

$$K \geq \frac{\frac{1}{2}\log(1/\rho) - \log 440}{\log 16} - 3.$$

For all sufficiently small universal $\rho_0 > 0$, this ensures the cardinality required in (N1), $K \geq c_2 L$, and $\log K \geq c_3 H$. Equation (N7) then gives $P \geq c_4 \rho LH$. For $\rho \in [\rho_0, 1/4]$ use $P \geq 2\rho/5$ and boundedness of LH on that fixed compact interval. This proves the universal lower bound for MW. Finally (N4) yields the asserted bound for VW. □

Corollary 3.2. *Suppose an estimator is the deterministic arithmetic average of $n \geq 1$ independent copies of the same single-coin procedure, and its uniform expected work is at most B. If its worst-case RMS error is at most $\eta \leq \rho/64$, then $B\eta^2 \geq c\rho LH$. The procedure and n may depend on B, ρ.*

Proof. For every q the squared error of the average is $\mathrm{Var}_q(Y)/n + (\mathbb{E}_q Y - \theta(q))^2$. Thus its kernel has bias at most η and $V/n \leq \eta^2$. Uniform expected work gives $nW \leq B$, so $B\eta^2 \geq (nW)(V/n) = VW \geq c\rho LH$. □

The last corollary does not cover clipping or any other nonlinear postprocessing, unequal kernels whose biases cancel, data-dependent replication counts, or interaction among genuine reservoir coins. In particular, a raw variance lower bound cannot be substituted for a lower bound on clipped probability-estimation risk. The unrestricted population minimax problem remains open in this pass.

4 A population information reduction and a radial obstruction

PROVED-ON-PAPER. These statements concern the actual iid hidden-gap experiment. They neither reveal a sampled gap nor provide an oracle for the unknown sign allocation. The reduction below is finite-prefix; the stated testing consequence uses almost-sure termination separately at the two endpoint laws. No unproved differentiability claim at an unbounded stopping time is used. No Lean formalization or priority claim is made.

4.1 Directional information inequalities tensorize over interleaving

Fix a known magnitude law on $(0, a]$, $a < 1/2$, and a path $q_t = q_* + tu$, $t \in [\alpha, \beta]$, of measurable feasible sign allocations in $\mathcal{Q}_\rho$. The algorithm is independent of t. When $\alpha < \beta$, feasibility at the two endpoints implies $|u| \le 1/(\beta - \alpha)$ almost everywhere; the degenerate path is trivial. For a finite bit word h write $f_t(h)$ for its probability on one fresh coin, and set

$$s_t(h) = \partial_t \log f_t(h), \quad p_t(b \mid h) = f_t(hb)/f_t(h), \quad v_t(h) = \sum_{b=0}^{1} p_t(b \mid h)[s_t(hb) - s_t(h)]^2.$$

Here $f_t(h)$ is affine in t and strictly positive, because all Bernoulli parameters lie in $[1/2 - a, 1/2 + a]$. A bounded number of observations therefore has a well-defined ordinary Fisher information. Include the independent random seed in each transcript; its law is fixed when taking a derivative.

Lemma 4.1 (Finite-prefix information/work direct sum)**.** *Suppose a measurable function $\kappa(t) \ge 0$ has this property: for each fixed t, every randomized single-coin policy with a deterministic finite cap on its number of flips satisfies*

$$I_t(\textit{seed and stopped coin history}) = \mathbb{E}_t \sum_{k<\tau} v_t(H_k) \le \kappa(t)\mathbb{E}_t\tau.$$

Then every population algorithm with at most $N < \infty$ total flips, arbitrary interleaving and adaptive creation of iid coins satisfies

$$I_t(\textit{full population transcript}) = \mathbb{E}_t \sum_{n<T} v_t(H_{i_n,n}) \le \kappa(t)\mathbb{E}_t T.$$

Here $H_{i_n,n}$ is the previous local history of the coin selected for the next observation. The single-coin premise may grant its policy knowledge of the reference value t; the derivative keeps that policy fixed.

Proof. Conditioning on the seed, each possible transcript has likelihood equal to the product of $f_t(h_i)$ over the local histories of its distinct coins. The adaptive scheduling and stopping choices are already determined by the displayed seed and past observations, so contribute no likelihood factor depending on t. Revealing the next bit changes the score by $s_t(hb) - s_t(h)$, which has conditional mean zero. Squaring the finite sum of score increments and taking expectations makes the cross terms vanish. This proves the two information identities.

For the inequality fix a reference t and a creation index $i \le N$, where coins are indexed by their first charged observation. Run a copy of the capped population policy with its ith such coin supplied by one genuine coin, and simulate all other coins using independent tapes drawn from the known reference law q_t. The simulated tapes are part of an independent seed. They are held fixed in law while differentiating the genuine coin's distribution. If the ith coin is never requested, take zero genuine flips. This is a valid one-coin policy using at most N genuine flips; simulated observations and arithmetic are only auxiliary randomization in this information comparison. Its expected information and work at the reference law are exactly the contributions of coin i to the two displayed population sums. The hypothesis bounds that contribution by $\kappa(t)\mathbb{E}_t T_i$. Summing over $i \le N$ proves the result. No simulated reference coins are supplied to the original statistical algorithm. □

The proof is not specific to Bernoulli observations: it applies to any independent reservoir observation channel having positive differentiable finite-history likelihoods, provided differentiation under the seed and observation integrals is justified and the stated one-coin information/work premise holds. This is a conditional transfer principle, not a guessed universal information functional for other channels or cost models.

Corollary 4.2 (Expected-budget testing consequence)**.** *Let an unrestricted population procedure satisfy* $\sup_{t\in[\alpha,\beta]} \mathbb{E}_t T \le B < \infty$*. If the premise of Lemma 4.1 holds, then the total variation distance between its two terminal transcript laws is at most*

$$\frac{\sqrt{B}}{2}\int_\alpha^\beta \sqrt{\kappa(t)}\,dt.$$

Consequently, if this quantity is at most $1/2$ *and* $\Delta = |\theta(q_\beta)-\theta(q_\alpha)|$*, its worst-case RMS error over these two laws is at least* $\Delta/4$*.*

Proof. For every finite prefix capped at N flips, integrate its score derivative along the path and apply Cauchy--Schwarz:

$$\|P_{\beta,N}-P_{\alpha,N}\|_{\mathrm{TV}} \le \tfrac{1}{2}\int_\alpha^\beta \sqrt{I_t(P_{t,N})}\,dt \le \tfrac{\sqrt{B}}{2}\int_\alpha^\beta \sqrt{\kappa(t)}\,dt.$$

Finite prefix densities are finite sums over observation/action branches conditional on the seed, so differentiating is legitimate; the same identity extends by integration over the fixed independent seed. Finite expected work implies almost-sure termination separately at each endpoint. For any terminal event E, its intersection with $\{T \le N\}$ is determined by that prefix. Thus its endpoint probability difference is bounded by the preceding total-variation bound plus $\mathbb{P}_\alpha(T > N) + \mathbb{P}_\beta(T > N)$, which tends to zero. Take the supremum over E. Finally threshold an estimator at the midpoint of the two target values. The sum of its two testing errors is at least $1 - \mathrm{TV} \ge 1/2$, whereas Markov's inequality bounds that sum by $8R^2/\Delta^2$. Hence $R \ge \Delta/4$. □

The missing hypothesis is an upper bound on *directional Fisher information per expected flip for every single-coin policy*. A lower bound on the uniform variance--work product of estimators of the entire functional does not provide this hypothesis. The next example shows why the most natural radial path cannot supply the extra critical logarithm.

4.2 The radial path has too much information for that converse

Fix again $a = 1/4$, the exact quadratic law, and define $b \in (0, a)$ by

$$w_b(d) = \min\{1, b^2/d^2\}, \qquad \mathbb{E}w_b(D) = \rho.$$

Set $\ell = \log(a/b)$ and $L = 1+\log(1/\rho)$. Then $\rho = (b/a)^2(1+2\ell)$ and $L = 1+2\ell-\log(1+2\ell)$. Consider the genuine one-dimensional submodel

$$q_\eta(d) = \tfrac{1}{2}(1+\eta)w_b(d), \qquad -1/2 \le \eta \le 1/2.$$

It belongs to $\mathcal{Q}_\rho$, and its target derivative is $\rho/2$.

Proposition 4.3. *At* $\eta = 0$*, the supremum of single-coin information/work ratios over finite-cap policies is at least*

$$\frac{a^2\rho}{32L}.$$

Thus no uniform upper bound of order $a^2\rho/[L(1+\log L)]$ *can hold for this directional information envelope as* $\rho \downarrow 0$*.*

Proof. Let $M = \lceil 8a/b \rceil$. On the genuine coin, run the integer walk with increments $2X-1$ from zero until it hits -1 or M; output the indicator A of hitting M. Let $\pi_\pm(d)$ be its success probabilities conditional on mean $1/2 \pm d$. The elementary gambler's-ruin formula gives

$$\pi_+(d) = \frac{1-r}{1-r^{M+1}}, \quad r = \frac{1-2d}{1+2d}, \qquad \pi_-(d) = \frac{R-1}{R^{M+1}-1}, \quad R = r^{-1}.$$

These follow by solving the two-boundary recurrence, or by summing its geometric first differences. In particular

$$\frac{8}{3}d \le \pi_+(d) \le 4d + \frac{1}{M+1}, \qquad \pi_-(d) \le \frac{8d}{e^{4(M+1)d} - 1}. \tag{N8}$$

For the upper positive bound, write $\pi_+ = (1-r) + r^{M+1}\pi_+$ and use $r^{M+1}(M+1) \le \sum_{k=0}^{M} r^k = 1/\pi_+$. The negative bound uses $R - 1 \le 8d$ and $\log R \ge 4d$.

Write $F = \mathbb{E}\pi_-(D)$ and $A_+ = \mathbb{E}[q_0(D)\pi_+(D)]$. Expanding $(e^x - 1)^{-1} = \sum_{k\ge1} e^{-kx}$ and using $\sum_{k\ge1} k^{-3} < 2$ yields

$$F \le \frac{16}{a^2}\int_0^\infty \frac{d^2}{e^{4(M+1)d} - 1}\, dd < \frac{1}{a^2(M+1)^3} \le \frac{b^3}{512a^5}.$$

Direct integration gives $\mathbb{E}[Dw_b(D)] = 2b^2/a - 4b^3/(3a^2) \ge 2b^2/(3a)$. Hence (N8) gives

$$A_+ \ge \tfrac{4}{3}\mathbb{E}[Dw_b(D)] \ge \frac{8}{9}\frac{b^2}{a}, \qquad \frac{F}{A_+} \le \frac{9b}{4096a^4} \le \frac{9}{64} < \frac{1}{4}.$$

Let $z_\eta = \mathbb{P}_\eta(A = 1)$. Because this single-coin law is affine in η,

$$z_0 \le A_+ + F \le \tfrac{5}{4}A_+, \qquad z_0' = \mathbb{E}[q_0(\pi_+ - \pi_-)] \ge A_+ - F \ge \tfrac{3}{4}A_+.$$

Both boundary events have strictly positive probability at every signed gap; hence $0 < z_0 < 1$. The information in the output indicator alone therefore satisfies

$$I_0(A) = \frac{(z_0')^2}{z_0(1-z_0)} \ge \frac{9}{20}A_+ \ge \frac{2}{5}\frac{b^2}{a}. \tag{N9}$$

For fixed signed d, the hitting time has finite mean: in every block of $M+1$ observations there is a positive, uniform-in-the-current-state chance of hitting a boundary, because both bit probabilities are at least $1/4$. The stopped walk identity (or the same finite difference recurrence) gives

$$\mathbb{E}_{-d}\tau = \frac{1 - (M+1)\pi_-(d)}{2d} \le \frac{1}{2d}, \qquad \mathbb{E}_{+d}\tau = \frac{(M+1)\pi_+(d) - 1}{2d} \le 2(M+1) + \frac{1}{2d}.$$

At $\eta = 0$, integration and $\mathbb{E}[w_b(D)/D] \le 4b/a^2$ imply

$$\begin{aligned}\mathbb{E}_0\tau &\le \frac{1}{a} + (M+1)\rho + \frac{b}{a^2} \\ &\le 8 + 8(b/a)(1+2\ell) + 2\rho < 25.\end{aligned}$$

The last inequality uses $e^{-\ell}(1+2\ell) \le 2$ for $\ell \ge 0$, which follows from $e^\ell \ge 1 + \ell$. Now set $u = 1 + 2\ell$. The elementary inequality $\log u \le u/2$ gives $L = u - \log u \ge u/2$, so $b^2 \ge a^2\rho/(2L)$. Combining this with (N9) and $a = 1/4$ shows

$$\frac{I_0(A)}{\mathbb{E}_0\tau} \ge \frac{2b^2}{125a} \ge \frac{a\rho}{125L} = \frac{a^2\rho}{(125/4)L} \ge \frac{a^2\rho}{32L}.$$

To obtain the asserted finite-cap result, stop after at most N flips and output one only if the upper boundary was already hit. For $N \ge M$ this event and its complement both have positive probability. Its success probability and its η derivative converge to z_0, z_0' by dominated convergence: the mixture probabilities are affine in η and $|q_\eta'| = q_0 \le 1/2$. Its work increases to $\mathbb{E}_0\tau$. The finite-cap output information therefore converges to $I_0(A)$, and the full transcript has at least this information. Taking the supremum over finite caps proves the proposition. □

4.3 An estimator at the candidate order on the known radial submodel

Corollary 4.4. *Under the known radial submodel of Proposition 4.3, and the ideal arithmetic convention allowing exact evaluation of known one-dimensional calibration integrals, there is an observable estimator with uniform expected work at most $B > 0$ and worst-case RMS at most*

$$\min\left\{\rho/4, \sqrt{160\rho L/B}\right\}.$$

The guarantee is only over q_η, $|\eta| \le 1/2$, and not over arbitrary sign allocations in $\mathcal{Q}_\rho$.

Proof. All expectations in the following calibration constants are integrals against the known exact quadratic law. Set

$$d_0 = \mathbb{E}[q_0(D)(\pi_+(D) - \pi_-(D))], \qquad F = \mathbb{E}\pi_-(D), \qquad Y = \frac{\rho}{2d_0}(A - F).$$

These constants depend only on the known submodel, not on η or on the realized gap. The previous proof gives $d_0 \ge (2/3)b^2/a > 0$ and $F \le d_0/3$. Since $z_\eta = F + (1+\eta)d_0$, $\mathbb{E}_\eta Y = (1+\eta)\rho/2 = \theta(q_\eta)$, and

$$\mathrm{Var}_\eta(Y) \le \frac{\rho^2}{4d_0^2} z_\eta \le \frac{11\rho^2}{24 d_0} \le \frac{11 a \rho^2}{16 b^2} \le \frac{11}{8a}\rho L = \frac{11}{2}\rho L.$$

The upper bound for π_+ in (N8), retaining the subtraction of one in the work formula, improves the positive conditional work bound to $\mathbb{E}_{+d}\tau \le 2(M+1)$. Thus uniformly in this submodel,

$$\mathbb{E}_\eta \tau \le \frac{1}{a} + 2(M+1)\theta(q_\eta) \le 4 + \frac{3}{2}(M+1)\rho \le 4 + \frac{3}{2}(16 + 1/2) < 29.$$

Each test terminates almost surely by the finite-interval argument above.

Independently draw $N \sim \mathrm{Poisson}(B/29)$ and run N independent copies on genuinely fresh coins, with independent seeds. Put $\lambda = B/29$ and output

$$\widehat{\theta} = \rho/2 + \lambda^{-1} \sum_{i=1}^{N} (Y_i - \rho/2).$$

This is a finite sum almost surely and costs at most $29\mathbb{E}N = B$ in expectation, uniformly in η. The compound-Poisson identities (obtained directly by conditioning on N) give $\mathbb{E}\widehat{\theta} = \theta$ and

$$\mathrm{Var}(\widehat{\theta}) = \lambda^{-1}\mathbb{E}_\eta(Y - \rho/2)^2 \le \frac{29}{B}\left(\frac{11}{2}\rho L + \frac{\rho^2}{16}\right) < \frac{160\rho L}{B},$$

using $\rho^2/16 \le \rho L/64$. The deterministic output $\rho/2$ has worst-case error $\rho/4$ on this submodel. Choose the smaller stated guarantee using known B, ρ. □

Calibration and practical scope. This is a proved estimator in the stated ideal computation model, not a held-out benchmark or claim of efficient numerical calibration. Its one-dimensional integrands are explicit and continuous, using the fair limit $\pi_\pm(0) = 1/(M+1)$, and the cutoff b is the unique root of a known strictly increasing continuous function. For a fixed justified integer M and a fixed known radial profile, these calibration integrals can be approximated to arbitrary certified precision. The following error bound holds with b, M fixed and approximates only F, d_0; it does not certify root-finding for b or rounding $\lceil 8a/b \rceil$. No such executed certification is claimed. For explicit error accounting, if $|\widehat{d_0} - d_0| \le d_0/2$ and $\widehat{Y} = \rho(A - \widehat{F})/(2\widehat{d_0})$, then pointwise

$$|\widehat{Y} - Y| \le \rho\left(\frac{|\widehat{d_0} - d_0|}{d_0^2} + \frac{|\widehat{F} - F|}{d_0}\right) =: \zeta.$$

Using the same independent Poisson count, the RMS difference of the calibrated and ideal final outputs is at most $\zeta\sqrt{1 + 1/\lambda}$, because it is pointwise at most $\zeta N/\lambda$. Add this term to the ideal RMS guarantee; flip costs are unchanged. No actual numerical integration or implementation run is claimed here, and any computational calibration cost must be reported separately in a practical comparison.

Pseudocode. Compute b, M, F, d_0 from the known radial model and select the better of the trivial radius and the displayed Poisson bound. If selecting the trivial bound, output $\rho/2$ immediately. Otherwise sample the independent Poisson count. For each of its genuinely fresh coins, accumulate $2X - 1$ until the running sum hits -1 or M, record the upper-hit indicator, and convert it to Y by the displayed calibration. Return the centered Poisson sum. No posterior involving the unknown η, no true gap, and no selected-magnitude request is used.

The proposition disproves a particular proposed directional information upper bound, not the unrestricted minimax rate. The calibrated estimator knows the radial profile w_b through the specified submodel and estimates $\int q\, d\mu$ only on that submodel. It charges every genuine flip and never uses the realized magnitude. The full nonparametric problem could still require an entropy loss arising from multiple nuisance directions, or could admit a nonlinear population construction absent from the single-coin product argument.

5 A separate restriction on independent pilot correction

Evidence and scope. The statements below are proved on paper, with exact rational finite checks in the accompanying script. They have no Lean formal counterpart and no external peer review. No novelty claim is made. They concern a restricted class of population procedures and do not resolve unrestricted minimax risk.

Experiment. Let $\mu = \sum_{i=1}^{k} \pi_i \delta_{d_i}$, where $\pi_i > 0$, $\sum_i \pi_i = 1$, and $0 < d_i \le 1/2$ are known. A new coin independently draws $D \sim \mu$ and $S \mid D = d_i \sim \mathrm{Bernoulli}(q_i)$, then supplies conditionally iid flips with success probability $1/2 + (2S-1)D$. The gap and sign are hidden. For known $0 < \rho \le 1/4$, the parameter space is the compact convex set

$$Q = \{q \in [0,1]^k : \ \theta(q) := \textstyle\sum_i \pi_i q_i \le \rho\}.$$

All coin observations, including pilot observations, are charged. Arithmetic and internal randomization are free. The three-flip construction below has a deterministic work count; no random-work approximation is used.

Theorem 5.1 (A saddle point for a fixed scalar control variate)**.** *Fix a single-coin observation rule and real features Y, Z of its observations and internal randomization. Suppose they have finite second moments under each conditional law $(D, S) = (d_i, s)$, and for all $q \in Q$,*

$$\mathbb{E}_q Y = \theta(q), \qquad \mathbb{E}_q Z = 0, \qquad \mathbb{E}_q Z^2 = v > 0,$$

where v does not depend on q. Put $f(q,\beta) = \mathrm{Var}_q(Y - \beta Z)$ and $F(\beta) = \max_{q \in Q} f(q,\beta)$. There exist $\beta_* \in \mathbb{R}$ and $q_* \in Q$ such that

$$f(q_*, \beta) \ \ge \ f(q_*, \beta_*) = F(\beta_*) = \min_{b \in \mathbb{R}} F(b) \quad \textit{for every } \beta \in \mathbb{R}.$$

In particular,

$$\sup_{q \in Q} \inf_{\beta \in \mathbb{R}} f(q,\beta) = \inf_{\beta \in \mathbb{R}} \sup_{q \in Q} f(q,\beta).$$

Proof. Write $c(q) = \mathbb{E}_q[YZ]$ and $V(q) = \mathrm{Var}_q Y$. Conditional mixture probabilities are affine functions of q. Therefore c is affine and continuous, $V(q) = \mathbb{E}_q Y^2 - \theta(q)^2$ is continuous and concave, and

$$f(q,\beta) = V(q) - 2\beta c(q) + v\beta^2$$

is concave in q and strictly convex in β. On compact Q, let $C = \max_q |c(q)| < \infty$. Since $V(q) \ge 0$, $F(\beta) \ge v\beta^2 - 2C|\beta|$, so F is coercive. It is continuous, being a maximum over a compact set of functions whose joint formula is continuous. It has a global minimizer β_*.

Let $\mathcal{A} = \{q : f(q, \beta_*) = F(\beta_*)\}$ be its nonempty compact active set. We show that the active slopes $g(q) = 2v\beta_* - 2c(q)$ include one nonpositive and one nonnegative slope. If every active slope were positive, compactness would give a neighborhood U of $\mathcal{A}$ on which $g \geq \eta > 0$. On the compact complement $Q \setminus U$, the value at β_* is at most $F(\beta_*) - \gamma$ for some $\gamma > 0$ (if the complement is empty, omit it). For sufficiently small $t > 0$, the identity

$$f(q, \beta_* - t) = f(q, \beta_*) - tg(q) + vt^2$$

then makes every value strictly smaller than $F(\beta_*)$: on U use $t\eta - vt^2 > 0$, and on its complement use the uniform bound on $|g|$ and the gap γ. This contradicts minimality. The all-negative case follows by increasing β_*. Thus there are $q_1, q_2 \in \mathcal{A}$ and $\lambda \in [0, 1]$ for which $\lambda g(q_1) + (1 - \lambda)g(q_2) = 0$.

Set $q_* = \lambda q_1 + (1 - \lambda)q_2 \in Q$. Concavity gives

$$f(q_*, \beta_*) \geq \lambda f(q_1, \beta_*) + (1 - \lambda) f(q_2, \beta_*) = F(\beta_*).$$

The opposite inequality is the definition of F. Moreover $g(q_*) = 0$ by affinity of c, so β_* minimizes the quadratic $f(q_*, \cdot)$. This proves the saddle assertion and both directions of the equality. □

Theorem 5.2 (Independent pilot learning does not improve this minimax risk)**.** *Under Theorem 5.1, take $n \geq 1$ independent fresh production coins, obtaining iid pairs (Y_i, Z_i). A pilot may use arbitrary additional coins and observations to choose an almost surely finite real coefficient $\widehat{\beta}$, but its complete transcript and randomization must be independent of the production coins under each q. Let*

$$\widehat{\theta} = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \widehat{\beta} Z_i), \qquad F_* := F(\beta_*).$$

Then, allowing infinite mean squared error,

$$\sup_{q \in Q} \mathbb{E}_q (\widehat{\theta} - \theta(q))^2 \geq F_*/n.$$

The fixed coefficient β_* attains equality without any pilot. If the features use exactly ℓ flips per coin and a pilot spends ℓm flips, then a fixed-coefficient estimate from $m + n$ coins instead has worst-case MSE $F_*/(m + n)$ at the same deterministic total work.

Proof. For each realized finite coefficient, the production estimate has conditional mean $\theta(q)$ and conditional squared error $f(q, \widehat{\beta})/n$. Taking expectations of this nonnegative conditional second moment is valid even when it is infinite. At the specific q_* in Theorem 5.1, every possible coefficient has $f(q_*, \widehat{\beta}) \geq F_*$. A fixed β_* has worst-case variance F_* per production coin and is unbiased. The work comparison follows by counting coins. The last assertion requires that the stated pilot work is indeed deterministic; no conversion from expected work is implicit. □

Excluded procedures. The final estimate here is the raw affine average. Clipping it, changing the production feature or sampling rule after seeing a pilot, adaptively selecting the production sample size, accepting bias, and nonlinear aggregation are all outside the theorem. Thus this result does not transfer a single-coin lower bound to unrestricted population procedures.

Proposition 5.3 (Explicit two-gap three-flip construction)**.** *Suppose $k = 2$, $0 < d_1 < d_2 \leq 1/2$, and put $x_i = 2d_i$. Take exactly three flips on a fresh coin and put $R_j = 2X_j - 1$,*

$$A = (R_1 + R_2 + R_3)/3, \quad B = R_1 R_2 R_3, \quad C = (R_1 R_2 + R_1 R_3 + R_2 R_3)/3.$$

Define

$$a = \frac{x_1^2 + x_1 x_2 + x_2^2}{2x_1 x_2 (x_1 + x_2)}, \qquad b = -\frac{1}{2x_1 x_2 (x_1 + x_2)}, \qquad m_2 = \pi_1 x_1^2 + \pi_2 x_2^2,$$

and $Y = 1/2 + aA + bB$, $Z = C - m_2$. *Then* $\mathbb{E}_q Y = \theta(q)$ *and* $\mathbb{E}_q Z = 0$ *for every* q. *Moreover*

$$v := \mathbb{E}_q Z^2 = (1 - m_2)(m_2 + 1/3) > 0.$$

Write $m_j = \sum_i \pi_i (2q_i - 1) x_i^j$ *for odd* j, *and*

$$K = a^2(1 + 2m_2)/3 + 2abm_2 + b^2.$$

The exact moments are

$$\begin{aligned} V(q) &= K - 1/4 + \theta(q) - \theta(q)^2, \\ c(q) &= a\{(2/3 - m_2)m_1 + m_3/3\} + b(m_1 - m_2 m_3). \end{aligned}$$

Consequently these features satisfy Theorem 5.1.

Proof. Conditionally on $D = d_i, S = s$, the R_j are iid with mean $x = (2s-1)x_i$. Hence $\mathbb{E}A = x$, $\mathbb{E}B = x^3$, $\mathbb{E}C = x^2$. The displayed coefficients solve $ax_i + bx_i^3 = 1/2$ for both i, so $\mathbb{E}[Y \mid D = d_i, S = s] = s$. The identity $\mathbb{E}[C \mid D = d_i, S = s] = x_i^2$ proves the assertion for Z. Direct multiplication using $R_j^2 = 1$ gives

$$\mathbb{E}A^2 = \mathbb{E}C^2 = (1 + 2m_2)/3, \quad \mathbb{E}B^2 = 1, \quad \mathbb{E}AB = m_2,$$

and

$$\mathbb{E}AC = 2m_1/3 + m_3/3, \qquad \mathbb{E}BC = m_1.$$

These yield the claimed variance and covariance formulas. Since the two gaps are distinct and both masses are positive, $0 < m_2 < 1$, even when $d_2 = 1/2$. Thus $v > 0$. All features are bounded for the fixed positive gaps, and no sampled latent value is used by the computation. □

Proposition 5.4 (A fully specified pilot implementation)**.** *Use* $m \geq 1$ *independent pilot coins with the features in Proposition 5.3. Set*

$$\widehat{\beta} = \frac{1}{mv} \sum_{j=1}^{m} Y_j Z_j.$$

On $n \geq 1$ *independent production coins return the raw average of* $Y - \widehat{\beta} Z$. *It is unbiased for all* q, *terminates after exactly* $3(m + n)$ *flips, and has exact MSE*

$$\frac{1}{n}\left(V(q) - \frac{c(q)^2}{v} + \frac{\mathrm{Var}_q(YZ)}{mv}\right).$$

Proof. The pilot is independent of production, with $\mathbb{E}\widehat{\beta} = c(q)/v$ and $\mathrm{Var}(\widehat{\beta}) = \mathrm{Var}_q(YZ)/(mv^2)$. Completing the square in $f(q, \beta) = V(q) - c(q)^2/v + v(\beta - c(q)/v)^2$ and then averaging proves the formula. Conditional unbiasedness proves unconditional unbiasedness. All moments are finite for fixed gaps, and the work count is deterministic. □

Interpretation. The explicit estimator is an exact hidden-gap construction, not an efficient rare-event algorithm: its interpolation coefficients grow as the gaps shrink, and it can have a large variance even at $\theta = 0$. The no-gain theorem shows why merely learning a coefficient for fixed unbiased features cannot be the sought minimax improvement. It leaves acquisition adaptation and nonlinear estimators as separate questions. No claim is made that the control-variate idea or the convex saddle argument is new.

6 Exact finite-horizon evidence

Two preregistered synthetic laws have respectively masses $(1/8, 7/8)$ at gaps $(1/32, 1/4)$ and masses $(1/16, 3/16, 3/4)$ at gaps $(1/32, 1/8, 1/2)$. In both cases $\rho = 1/8$. The finite prior is uniform over $q = 0$, feasible single-atom allocations with target $\rho/2$ or ρ, and $q \equiv \rho$, with duplicates removed. The prior draws one common q before generating any coins. Neither sampled signs nor gaps are observed.

At deterministic horizons one through eight we solve the finite Bayes allocation problem exactly with rational arithmetic. A state is the sorted list of heads/tails counts for each observed coin. Transcript likelihoods factor by coin; the terminal posterior mean is optimal, and minimizing the two-child expected posterior variance over fresh/revisited choices gives the Bellman recursion. Randomization and earlier stopping cannot improve the Bayes minimum. A separate labeled ordered-tape recursion agrees exactly at all ten case/horizon pairs through horizon five.

Law, horizon 8	Adaptive MSE	Best restricted MSE	Reduction
Two scales	0.002111783	0.002145941	1.5917%
Three scales	0.001706729	0.001797138	5.0307%

Restricted choices are one coin only or fresh coins only, with the same posterior-optimal terminal estimator. These are rounded displays of exact fractions, not estimates with sampling uncertainty. They concern a finite prior and a deterministic cap, not the uniform expected-budget minimax problem. They are not held-out practical performance claims.

7 Exact remaining obligations

A full resolution needs either a biased or nonlinear population estimator with uniform error and expected work at the candidate scale for every global allocation, or a population lower bound exploiting nuisance directions beyond the radial path. The current product lower bound does not apply after clipping or nonlinear aggregation. The pilot theorem fixes its production features and excludes adaptive acquisition. The information reduction supplies no automatic bound on its one-coin directional premise. Finite computations do not close any of these logical gaps. Extending the result to arbitrary measures, joint rarity/accuracy/confidence limits, or unknown-law adaptation remains separate work.

A Preserved inherited proof and upper checkpoint

This appendix is a packaging derivative of the byte-preserved concurrent source. Its document wrapper, title command, and appendix numbering command were removed; line breaks were allowed in source paths and hashes. Its mathematical content and manually tagged equations are unchanged. Its repeated status descriptions refer to that inherited work; the new review is recorded in the current audit.

Status and main limitation. This is a paper proof checked in separate agent reviews, not a Lean-checked or externally specialist-reviewed result. The review records accompany this note; model diversity is not guaranteed. It gives matching characterizations for unbiased and uniformly small-bias *single-coin* estimators, including all moment--work tradeoffs in those classes. It does not prove a lower bound for arbitrary population algorithms and does not refute the critical minimax conjecture. The latter remains open in this attempt.

B Exact scope and strongest statement

Fix $a = 1/4$ and the probability law

$$\mu(dd) = \frac{2d}{a^2}\mathbf{1}_{(0,a]}(d)\,dd.$$

The choice of one fixed interior a suffices for the obstruction. Let $0 < \rho \le 1/4$ and $\mathcal{Q}_\rho = \{q : 0 \le q \le 1,\ \int q\, d\mu \le \rho\}$, with all functions measurable. A fresh coin has hidden magnitude $D \sim \mu$, hidden sign $S \mid D = d \sim \mathrm{Ber}(q(d))$, and iid Bernoulli flips of parameter $1/2 + (2S-1)D$ conditional on (D, S).

A single-coin procedure A may know μ, a, ρ and may use arbitrary independent randomization. It observes the flips of this one coin only, stops at a possibly unbounded observable stopping time $T \in \mathbb{N}_0 \cup \{\infty\}$, and returns a real observable output Y. It may stop before the first flip. It cannot sample additional genuine reservoir coins; free parameter-independent simulated variables can be included in its seed. Neither D, S, nor q is observed. Assume that for every $q \in \mathcal{Q}_\rho$,

$$\mathbb{E}_q Y = \theta(q) := \int q\, d\mu, \qquad V = \sup_{q \in \mathcal{Q}_\rho} \mathrm{Var}_q\, Y < \infty, \qquad W = \sup_{q \in \mathcal{Q}_\rho} \mathbb{E}_q T < \infty.$$

Write $M = \sup_q \mathbb{E}_q Y^2$. This is finite because $M \le V + \rho^2$. All suprema in this note are over the stated nonempty class. Nontermination is recorded as $T = \infty$, so finite expected work implies almost-sure termination under every admissible population. The output can be defined to be zero on the resulting null nontermination event.

Theorem B.1 (Sharp critical single-coin product)**.** *There exist finite universal constants $0 < c < C$ such that for every $0 < \rho \le 1/4$, with $L = 1 + \log(1/\rho)$,*

$$c\rho L(1 + \log L) \le \inf_A V(A)W(A) \le C\rho L(1 + \log L). \tag{1}$$

The infimum ranges over exactly the single-coin procedures above and permits dependence on ρ. The same assertion holds with $M(A)$ in place of $V(A)$. The lower bound holds for every admissible procedure; no threshold family, finite depth, or numerical grid restriction is imposed.

The next theorem strengthens the lower bound beyond exact unbiasedness while preserving Theorem B.1 as a separate statement.

Theorem B.2 (Small relative bias does not remove the entropy loss)**.** *Keep the same fixed law and single-coin observation rules. Replace exact unbiasedness by*

$$\sup_{q \in \mathcal{Q}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64,$$

and retain separate finite uniform variance V and expected work W. There are universal $0 < c < C < \infty$ such that, for every $0 < \rho \le 1/4$,

$$c\rho L(1 + \log L) \le \inf_A V(A)W(A) \le C\rho L(1 + \log L). \tag{1b}$$

The same assertion holds with $M(A) = \sup_q \mathbb{E}_q Y^2$ in place of $V(A)$. The infimum in (1b) includes every single-coin policy meeting this bias bound, not just the unbiased subclass or any chosen classifier family.

The upper bound in (1) reuses the current separately reviewed paper proof in `research/2026-10-27-minimax-audit/theorem/critical.tex`, which supplies $M \le (165/56)\rho$ and $W \le 1350a^{-2}L(1 + \log L)$. That source is preserved, and its theorem is not claimed as new here. The new lower proof below is self-contained. Its finite-grid sign-testing argument follows the established entropy mechanism of Chen--Li--Qiao [1, Lemma 4.1 and Appendix D]; we prove the needed Bernoulli, separated-gap, one-sided-cost version explicitly rather than silently substituting their Gaussian formulation.

For the population minimax problem, the candidate modulus is $\Psi(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L}/(a\sqrt{B})\}$. Thus (1) shows that a proof through an unbiased single-coin variance--work bound of order ρL is impossible for this fixed law. Changing only the allocation between its variance and work cannot remove the iterated logarithm. A matching population algorithm would need a different argument or estimator class. No reduction from arbitrary biased or cross-coin algorithms to this class is asserted.

C A finite-grid sign-testing entropy bound

Lemma C.1. *There are universal $c_s > 0$ and an integer K_s such that the following holds. Let $0 < d_K < \cdots < d_1 \le 1/128$, with $d_j/d_{j+1} \ge 2$. One randomized observable procedure receives iid Bernoulli flips of unknown parameter $1/2 \pm d_j$, stops, and returns a sign. Suppose its error is at most $1/10$ at all $2K$ parameter values and its expected stopping time is finite there. Put*

$$c_j = d_j^2 \mathbb{E}_{-d_j}\tau, \qquad \bar{c} = K^{-1}\sum_{j=1}^{K} c_j.$$

For $K \ge K_s$,

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j}\tau \ge c_s K \log K. \tag{2}$$

Correctness and termination at the fair coin are not assumed.

Proof. Set $b_0 = 1/1024$, $n_j = \lfloor b_0/d_j^2 \rfloor$, and $u_j = \lceil 4\mathbb{E}_{-d_j}\tau \rceil$. Since $d_j \le 1/128$, $n_j \ge b_0/(2d_j^2) \ge 1$. The event of returning minus by time n_j is measurable in the independent seed and the first n_j flips, including unused flips if the procedure stops earlier. Between the laws $1/2 - d_j$ and $1/2 + d_j$, the KL divergence of these full finite prefixes is at most

$$n_j \frac{(2d_j)^2}{(1/2 + d_j)(1/2 - d_j)} \le (64/3) n_j d_j^2 \le 1/48.$$

Here Bernoulli KL is at most its chi-square divergence. Pinsker bounds total variation by less than $1/8$. Since the event is an error under the plus law, its probability under the minus law is at most $1/10 + 1/8$. Markov gives $\mathbb{P}_{-d_j}(\tau > u_j) \le 1/4$. Consequently the finite-window event

$$A_j = \{\text{output minus and } n_j < \tau \le u_j\}$$

has probability at least $1 - 1/10 - (1/10 + 1/8) - 1/4 > 1/4$ under the minus law.

Let P_0 denote the fair-coin experiment with the same independent seed and policy. At a finite prefix of length u_j, its KL from the minus law is at most $4d_j^2 u_j \le 16c_j + 1$. For any event of probability $p \ge 1/4$ under that minus law and probability z under P_0, binary data processing and $\mathrm{kl}(p, z) \ge p\log(1/z) - \log 2$ give

$$P_0(A_j) \ge \exp(-64c_j - 8). \tag{3}$$

All information comparisons here concern deterministic finite prefixes. There is no optional-stopping identity under P_0, which may fail to halt.

At least $K/2$ indices form a set G with $c_j \le 2\bar{c}$. For those indices,

$$(n_j, u_j] \subset [b_0/(2d_j^2), (8\bar{c} + 1)/d_j^2].$$

The numbers d_j^{-2} are separated by factors at least four. At every finite time, the number of these intervals containing that time is therefore at most

$$H(\bar{c}) = 1 + \log_4((16\bar{c} + 2)/b_0).$$

On every realization under the fair law, all events A_j concern the same terminal time and output. Summing their indicators, then taking expectations, gives $\sum_{j\in G} P_0(A_j) \le H(\bar{c})$, even on paths that never halt. Together with (3),

$$(K/2)e^{-128\bar{c}-8} \le 1 + \log_4(1024(16\bar{c} + 2)). \tag{4}$$

This entails $\bar{c} \ge (\log K)/512$ for all sufficiently large universal K. One can take $K_s = \lceil e^{32} \rceil$: if $\bar{c} < (\log K)/512$ and $x = \log K \ge 32$, the left side of (4) exceeds $\frac{1}{2}e^{3x/4-8} \ge \frac{1}{2}e^{x/2} \ge x^2/16$, while its right side is at most $1 + \log_4(32x + 2048) < 6 + x/4$. These inequalities contradict $x \ge 32$. Thus $c_s = 1/512$ suffices. □

D What unbiasedness for every global-rarity allocation forces

For a fixed procedure first define the nonnegative extended conditional moments

$$M_s(d) = \mathbb{E}_{d,s} Y^2, \quad w_s(d) = \mathbb{E}_{d,s} T.$$

These are measurable nonnegative conditional moment functions, defined by the explicit iid Bernoulli kernel and independent seed. Their integrability is proved before signed manipulations: the admissible choices $q = 0$ and $q \equiv \rho$ imply $\int M_- d\mu \le M$ and $\int M_+ d\mu \le M/\rho$, as well as finite integrated work on both signs. All conditional moments used below are thus finite outside one common μ-null set. On the set $\{M_s(d) < \infty\}$ define the integrable signed conditional mean $m_s(d) = \mathbb{E}_{d,s} Y$, and assign it zero on its exceptional null set. This avoids interpreting any undefined difference of infinite integrals.

Unbiasedness at $q = 0$ gives $\int m_- d\mu = 0$. For every measurable E of positive mass, take $q = \min(1, \rho/\mu(E))\mathbf{1}_E$. Subtracting its unbiasedness identity from the one at zero proves $\int_E (m_+ - m_-) d\mu = \mu(E)$. Hence

$$m_+(d) - m_-(d) = 1 \quad \mu\text{-almost everywhere.} \tag{5}$$

This uses the global constraint only; it does not assume $q \le \rho$ pointwise. Conditional Jensen at the constant prior $q \equiv \rho$ now gives

$$\mathbb{E}_\rho Y^2 \ge \int [(1-\rho) m_-^2 + \rho(m_- + 1)^2] d\mu = \int m_-^2 d\mu + \rho \ge \rho.$$

Consequently

$$M \ge \rho, \qquad V \ge \rho(1-\rho), \qquad M \le V + \rho^2 \le \tfrac{4}{3} V. \tag{6}$$

The response identity is stronger than a single unbiasedness statement at a known mixture: its pointwise content will make a sign-testing reduction possible.

E Removing artificial thinning and controlling all product allocations

The event $A = \{T \ge 1\}$ is determined by the independent seed before any observation. Its probability p is independent of every latent state. It is positive, since otherwise the expectation of Y could not vary with q. Put $c = \mathbb{E}[Y \mathbf{1}_{A^c}]$, also independent of q. Use the conditional seed law on A to define the active procedure

$$Z = pY + c \quad \text{conditional on } A, \qquad T_Z = T \quad \text{conditional on } A.$$

This is a valid mathematical single-coin procedure. Conditioning can be implemented by resampling independent seeds until A occurs, before making any charged observation. The constants p, c are properties of the original policy, independent of D, S, q; the reduction may give them to the test. The active procedure always makes at least one flip, and

$$\mathbb{E}_q Z = \theta(q), \qquad W_Z = W/p \ge 1.$$

For $p < 1$, Cauchy--Schwarz on A^c and completing the square give

$$\begin{aligned} \mathbb{E}_q Z^2 &= p \mathbb{E}_q [Y^2 \mathbf{1}_A] + 2c\theta - c^2 \\ &\le pM + 2c\theta - c^2/(1-p) \le pM + (1-p)\theta^2 \le m := pM + \rho^2. \end{aligned}$$

For $p = 1$ take $c = 0$ and the same final bound holds directly. Applying (6) to Z shows $pM \ge \rho - (1-p)\rho^2 \ge \rho(1-\rho)$. Write $P = MW$. The resulting useful bounds are

$$\rho \le m \le \tfrac{4}{3} pM \le \tfrac{4}{3} P, \qquad mW_Z \le \tfrac{4}{3} P, \qquad P \ge \tfrac{3}{4}\rho. \tag{7}$$

Thus random refusal to observe the coin cannot evade the product argument.

F Good gaps and the same-coin sign reduction

Use conditional means and moments for Z in this section, denoted again by $m_s(d), M_s(d), w_s(d)$. We have $\sup_q \mathbb{E}_q Z^2 \le m$ and response identity (5). The set $B_- := \{M_- > 1/64\}$ has mass at most $64m$. The set

$$B_+ := \{M_+ - M_- > 2m/\rho\}$$

has mass less than ρ: otherwise nonatomicity of the exact power law gives a subset $E \subset B_+$ of mass ρ, and the admissible $q = \mathbf{1}_E$ would give $\mathbb{E}_q Z^2 = \int M_- d\mu + \int_E (M_+ - M_-) d\mu > 2m$, a contradiction. After excluding also the common null sets above, the good set G has complement of mass at most $64m + \rho \le 65m$. On G,

$$|m_-| \le 1/8, \qquad m_+ = m_- + 1 \ge 7/8, \qquad M_- \le 1/64, \qquad M_+ \le 1/64 + 2m/\rho. \tag{8}$$

Let $n = \lceil 256(1+m/\rho) \rceil \le 513 m/\rho$; the last inequality uses $m \ge \rho$. Run n independent restarts of the active procedure on the *same* fixed-gap coin, with independent seeds and consecutive unused flip blocks, and output plus iff their average exceeds $1/2$. The test is independent of which good gap is supplied. Conditional on its fixed latent state, each restart has the stated law, and the outputs are iid. To justify this with unbounded stopping, condition on each finite first stopping time and finite observed prefix; unused iid coordinates retain their original law. Sum over the countably many prefixes, whose probabilities sum to one because conditional expected work is finite. Repeating this argument proves the restart assertion and additivity of expected work. By (8), the threshold is separated from either mean by at least $3/8$. Chebyshev gives error at most $1/10$ on both signs, because

$$\frac{64}{9n} \max\{1/64, 1/64 + 2m/\rho\} < 1/10.$$

Its expected work under the minus sign at gap d is exactly $n w_-(d)$. This construction has not drawn additional latent gaps: it repeatedly observes one fixed coin, as required by Lemma C.1.

G Selecting geometric gaps from the continuous law

For integers $j \ge 3$ set

$$I_j = (a4^{-j}/2, a4^{-j}], \qquad \mu(I_j) = \tfrac{3}{4}\, 16^{-j}.$$

Their intervals are disjoint and any choices $d_j \in I_j$ satisfy $d_j/d_{j+1} > 2$; also $d_j \le a/64 = 1/256 < 1/128$. Let

$$J = \left\lfloor \log_{16} \frac{3}{520m} \right\rfloor, \qquad K = J - 2,$$

and consider the case $J \ge 3$. For $3 \le j \le J$, each annulus has mass at least $130m$, so $\mu(G \cap I_j) \ge \mu(I_j)/2$. Choose $d_j \in G \cap I_j$ with

$$w_-(d_j) \le \frac{4}{\mu(I_j)} \int_{I_j} w_-\, d\mu. \tag{9}$$

Such a choice follows from the average on $G \cap I_j$. The integral is strictly positive because the active procedure always makes at least one observation, so $w_-(d) \ge 1$. All exceptional null sets have already been removed. These are existential selections used only in the proof; the procedure is not given their values or the latent gap.

Since $d_j^2 \le a^2 16^{-j}$, summing (9) yields

$$\sum_{j=3}^{J} d_j^2 w_-(d_j) \le \frac{16a^2}{3} \int w_-\, d\mu \le \frac{16a^2}{3} W_Z.$$

Apply Lemma C.1 to the same test from the preceding section. For $K \geq K_s$, its cost lower bound and (7) give

$$c_s K \log K \leq \sum_{j=3}^{J} d_j^2 n w_-(d_j) \leq \frac{16a^2}{3} \frac{513m}{\rho} W_Z \leq \frac{3648a^2 P}{\rho}. \tag{10}$$

The coefficient $3648 = (16/3) \cdot 513 \cdot (4/3)$ is deliberately unoptimized. This proves a quantitative implicit product obstruction with every parameter and exceptional set accounted for.

H Completion of the product theorem and what it leaves open

We deduce an explicit order from (10) as $\rho \downarrow 0$, keeping $a = 1/4$. If $P \geq \sqrt{\rho}$, then $P \geq c\rho L(1 + \log L)$ for all sufficiently small ρ, since $\sqrt{\rho} L(1 + \log L) \to 0$. If $P < \sqrt{\rho}$, (7) gives $m < (4/3)\sqrt{\rho}$, and the definition of K implies

$$K \geq \frac{\frac{1}{2} \log(1/\rho) - \log(2080/9)}{\log 16} - 3.$$

For sufficiently small universal ρ, this is at least a universal positive multiple of L, exceeds K_s, and has $\log K \geq \frac{1}{2} \log L$. Equation (10) then gives $P \geq c\rho L(1+\log L)$, with a universal $c > 0$. All thresholds here are universal because a was fixed. For the remaining compact interval of rarities, reduce c if necessary and use $P \geq 3\rho/4$ from (7). Thus the raw-second-moment product lower bound holds for every $0 < \rho \leq 1/4$. Finally $M \leq (4/3)V$ by (6), so the same order lower bound holds for VW. The previously available upper construction supplies the other direction, proving Theorem B.1.

The result explains a real statistical loss in the unbiased single-coin route: its pointwise response identity permits recovery of a uniformly accurate sign test at many separated gap scales. The entropy cost of those scales forces the iterated logarithm. This is stronger than failure of the particular likelihood dictionary or a numerical estimator search, because the proof permits arbitrary single-coin observable policies and arbitrary separate variance--work allocations.

The result alone does not exclude biased estimators. The next section proves the separate extension stated in Theorem B.2. It remains possible that nonlinear use of population data or interaction between multiple genuine coins attains the candidate Ψ rate. The current proof neither constructs such an algorithm nor proves that it cannot exist. An extension would require a valid reduction from the full population minimax class to one of these single-coin classes, or a new direct information argument. Such a reduction is an open premise, not an accepted step of this note.

I Proof of the small-bias extension

Put $\eta = 1/64$. Consider a procedure with $\sup_q |\mathbb{E}_q Y - \theta(q)| \leq \eta\rho$ and the same finite separate bounds. As before, the nonnegative conditional moments are integrated first under $q = 0$ and $q \equiv \rho$, and the signed means are defined only where square integrable. Put $b(d) = \mathbb{E}_{d,-} Y$ and $f(d) = \mathbb{E}_{d,+} Y - \mathbb{E}_{d,-} Y$. The two population bias bounds give

$$\int f \, d\mu = \frac{\mathbb{E}_\rho Y - \mathbb{E}_0 Y}{\rho} \geq 31/32.$$

The law of total variance, conditioning first on D and then on S, gives

$$V \geq \mathrm{Var}_\rho(Y) \geq \rho(1-\rho) \int f^2 \, d\mu \geq \rho(1-\rho)(31/32)^2 \geq \tfrac{2}{3}\rho. \tag{14}$$

There is no pointwise response assumption in this step. Since every absolute population mean is at most $65\rho/64$, also

$$M \leq V + (65\rho/64)^2 \leq \tfrac{3}{2} V. \tag{15}$$

Normalize the first-observation event as before. Its probability p is positive: an output taking no observations has one common expectation and cannot approximate both $\theta = 0$ and $\theta = \rho$ within $\rho/64$.

For $c = \mathbb{E}[Y\mathbf{1}_{\{T=0\}}]$, use the active output $Z = pY + c$ conditional on $T \geq 1$. Its expectations are exactly those of Y, so the bias guarantee is unchanged, and $W_Z = W/p \geq 1$. The same completed-square computation, with the actual mean $\mathbb{E}_q Y$ in place of θ, yields

$$\sup_q \mathbb{E}_q Z^2 \leq pM + (65\rho/64)^2 \leq m := pM + 2\rho^2.$$

Applying (14) to Z shows $m \geq 2\rho/3$ and $pM \geq 2\rho/3 - 2\rho^2 \geq \rho/6$. Consequently, for $P = MW$,

$$m \leq 4pM \leq 4P, \qquad mW_Z \leq 4P, \qquad P \geq \rho/6. \tag{16}$$

Indeed $2\rho^2 \leq \rho/2 \leq 3pM$. Thus arbitrary pre-observation thinning and moment--work reallocations are again covered.

Use conditional moments for Z in what follows. The negative-moment bad set $\{M_- > 1/64\}$ has mass at most $64m$. As in the unbiased proof, $\{M_+ - M_- > 2m/\rho\}$ has mass less than ρ. The additional response bad set

$$B_f = \{d : f(d) < 3/4\}$$

also has mass less than ρ. If it had mass at least ρ, choose a measurable $E \subset B_f$ of mass ρ and the admissible $q = \mathbf{1}_E$. Then

$$\mathbb{E}_{\mathbf{1}_E} Z - \mathbb{E}_0 Z = \int_E f \, d\mu < 3\rho/4,$$

while the two bias bounds force this difference to be at least $\rho - 2\rho/64 = 31\rho/32$, a contradiction. This argument uses only global-rarity alternatives and does not assume pointwise classification errors.

After excluding these three bad sets and the common null sets, the good set has complement of mass at most $64m + 2\rho \leq 67m$. On it,

$$|m_-| \leq 1/8, \qquad m_+ \geq 5/8, \qquad M_+ \leq 1/64 + 2m/\rho, \qquad M_- \leq 1/64.$$

Restart Z independently $n = \lceil 512(1 + m/\rho) \rceil \leq 1282m/\rho$ times on the same fixed coin; the inequality uses $m/\rho \geq 2/3$. Threshold their average at $3/8$. Both mean separations are at least $1/4$, so Chebyshev bounds both errors by

$$\frac{16(1/64 + 2m/\rho)}{n} < 1/10.$$

The iid unused-suffix argument already proved gives expected work $nw_-(d)$ on a negative gap. No additional genuine coin or magnitude is introduced.

Use the same annuli I_j, now retaining

$$j \leq J = \left\lfloor \log_{16} \frac{3}{536m} \right\rfloor, \qquad K = J - 2.$$

Each retained annulus has mass at least $134m$, and the good set occupies at least half. The identical averaging selection and sign-entropy lemma give, for $K \geq K_s$,

$$c_s K \log K \leq \sum_j d_j^2 n w_-(d_j) \leq \frac{16a^2}{3} \frac{1282m}{\rho} W_Z \leq \frac{30000a^2 P}{\rho}. \tag{17}$$

Here $30000 > (16/3) \cdot 1282 \cdot 4$ is a conservative numerical constant. If $P < \sqrt{\rho}$, (16) shows $m \leq 4\sqrt{\rho}$, so K is at least a universal positive multiple of $\log(1/\rho)$ for sufficiently small ρ. Equation (17) proves the required order lower bound. If $P \geq \sqrt{\rho}$, that order already holds for small ρ; extend to the remaining compact interval using $P \geq \rho/6$. Finally (15) transfers the lower bound from MW to VW. The unbiased upper construction also belongs to this bias class, which proves both sides of Theorem B.2.

Exact averaging consequence. For an unprojected mean of a fixed number n of iid outputs of one such procedure, a uniform expected budget $nW \le B$ implies worst-case variance at least $V/n \ge VW/B$. Independent Poisson replication with mean count λ and output normalized by λ has variance $\mathbb{E}_q Y^2/\lambda$, giving the corresponding raw-moment product obstruction when $\lambda W \le B$. In particular small uniform per-coin bias does not remove this loss from either of these averaging arguments. These consequences do not include arbitrary nonlinear aggregation, data-dependent repetition counts, clipping, or interaction between genuine coins. No all-algorithm minimax conclusion follows without an additional valid reduction.

J Self-contained upper checkpoint and provenance

This appendix records the existing upper argument needed for the second inequality in (1), so the characterization does not depend on an inaccessible working-tree file. The inspected source was `research/2026-10-27-minimax-audit/theorem/critical.tex`, SHA-256

```
0e8b9eaf762b3b09fa0d92ce1a036c2c
eea8c12d00a5eea8286b852f36a81536
```

It was read without modifying it. The following is a compact restatement of that pre-existing proof, not a new construction or discovery claim. The source is now committed at `a8a8dc49115aabdd1fba8d53d3390eb0a5eee864`.

Put $H = 1 + \log L$. For $j \ge 0$ choose

$$\varepsilon_j = 16^{-(j+1)}, \qquad N_j = \lceil (a^2 \rho \varepsilon_j)^{-1} \rceil, \qquad K_j = \lceil \log_2(a\sqrt{N_j}) \rceil.$$

For $0 \le k \le K_j$, let $r_k = a2^{-(k+1)}$ and, after n observations with h_n heads, define

$$L^+_{k,n} = (1+2r_k)^{h_n}(1-2r_k)^{n-h_n}, \qquad L^-_{k,n} = (1-2r_k)^{h_n}(1+2r_k)^{n-h_n}.$$

Before N_j, stop with output zero if a negative likelihood reaches $(K_j+1)/\varepsilon_j$, or with output one if a positive likelihood reaches $(K_j+1)/(\rho\varepsilon_j)$. Fix zero priority for ties. At N_j, if no earlier decision was made, return the strict majority bit. Denote this bounded, observable classifier by Z_j and its work by T_j.

For the wrong-sign likelihood, each next multiplier has mean $1 - 4r_k d \le 1$. Stopping this nonnegative supermartingale at a crossing and a deterministic horizon, and then increasing the horizon, bounds the probability of crossing level b by $1/b$. Summing over K_j+1 references gives wrong-crossing probabilities at most ε_j on a positive coin and $\rho\varepsilon_j$ on a negative coin. The wrong forced-majority event is a subset of the full-tape event, of probability at most $e^{-2N_j d^2}$. This is Hoeffding's bound: the centered Bernoulli log moment-generating function has second derivative at most $1/4$, so is at most $\lambda^2/8$; use independence and exponential Markov with $\lambda = \pm 4d$. For the exact power law,

$$\mathbb{E}_\mu e^{-2ND^2} = \frac{1 - e^{-2Na^2}}{2Na^2} \le \frac{1}{2Na^2}.$$

No independence of a stopping event and this full-tape majority event is used. It follows, uniformly over the global-rarity class, that

$$\mathbb{P}(Z_j \ne S) \le \tfrac{5}{2}\rho\varepsilon_j, \qquad \mathbb{P}(Z_j = 1, S = 0) \le \tfrac{3}{2}\rho\varepsilon_j. \tag{11}$$

For $d \ge N_j^{-1/2}$, choose $k = \lceil \log_2(a/d) \rceil \le K_j$. Then $d/4 \le r_k \le d/2$, and the correct-sign log likelihood has mean increment at least $2d^2/3$: use

$$\tfrac{1}{2}\log(1 - 4r_k^2) + d\log\frac{1+2r_k}{1-2r_k} \ge -\tfrac{8}{3}r_k^2 + 4dr_k \ge \tfrac{2}{3}d^2.$$

Its upward increment is less than one. The bounded stopped-sum identity for the crossing time σ at a finite horizon n gives $g\mathbb{E}(\sigma \wedge n) \le A_{j,s}$, where

$$A_{j,s} = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s\log(1/\rho).$$

Monotone convergence proves finite expected crossing time and, including the cap and the trivial small-gap case,

$$\mathbb{E}(T_j \mid d, s) \le \min\{N_j, (3/2)A_{j,s}/d^2\}. \tag{12}$$

Write $c_0 = 3/2$. Direct integration of the negative contribution gives

$$\mathbb{E}\min\{N, c_0 A/D^2\} \le (c_0 A/a^2)[1 + \log_+(Na^2)].$$

For positives, layer cake and $\mathbb{E}[q\mathbf{1}_E] \le \min\{\rho, \mu(E)\}$ give

$$\mathbb{E}[q\min\{N, c_0 A/D^2\}] \le \int_0^\rho \min\{N, c_0 A/(a^2 u)\}\, du \le (c_0 A/a^2)[1 + \log_+(Na^2\rho)].$$

The exact integral is $N\rho$ when $Na^2\rho \le c_0 A$, and otherwise is $(c_0 A/a^2)[1 + \log(Na^2\rho/(c_0 A))]$. Thus this step also enforces the global, rather than pointwise, rarity condition.

Elementary bounds from the definitions are

$$\begin{aligned} N_j a^2 &\le 2/(\rho\varepsilon_j), & K_j + 1 &\le 5(L + j + 1), \\ A_{j,0} &\le 7H(j+1), & A_{j,1} &\le 7H(j+1) + L, \\ 1 + \log(N_j a^2) &\le 5L(j+1), & 1 + \log(N_j a^2 \rho) &\le 5(j+1). \end{aligned}$$

For example, $\log(L + j + 1) \le \log L + j + 1$ and $4\log 2 < 3$ prove the bound on $A_{j,0}$. Substitution in (12) and the two integrals gives

$$\mathbb{E}T_j \le (225/2)a^{-2}LH(j+1)^2. \tag{13}$$

Independently choose a level J with $p_j = 2^{-(j+1)}$. For $J = 0$ return $Y = Z_0/p_0$ on one fresh coin. For $J = j \ge 1$, run Z_j and Z_{j-1} on successive unused blocks of that same coin, set $\Delta_j = Z_j - Z_{j-1}$, and return $Y = \Delta_j/p_j$. The bounded stopping times make the iid suffix argument elementary, as in the restart argument above; both decisions share the same latent sign. A disagreement implies an error of one classifier. Hence (11) gives

$$\mathbb{E}Z_0^2 \le \tfrac{35}{32}\rho, \qquad \mathbb{E}\Delta_j^2 \le \tfrac{85}{32}\rho 16^{-j} \quad (j \ge 1).$$

Tonelli and the exact geometric sum now give

$$\mathbb{E}Y^2 \le \frac{35}{16}\rho + \frac{85}{16}\rho\sum_{j\ge1} 8^{-j} = \frac{165}{56}\rho.$$

The disagreement bounds also give absolute summability of expected increments, so signed expectations may be telescoped after integrability is established: $\mathbb{E}Y = \lim_j \mathbb{E}Z_j = \theta$. Work costs at most twice (13), and $\sum_j p_j(j+1)^2 = 6$, yielding

$$\mathbb{E}T_Y \le 1350a^{-2}LH.$$

All operations are on one observable coin with independent seed; the random level is unbounded but the separate expectations are finite. This completes the upper checkpoint used in (1).

References

[1] L. Chen, J. Li, and M. Qiao. *Towards Instance Optimal Bounds for Best Arm Identification*. COLT 2017; arXiv:1608.06031v2. `https://arxiv.org/abs/1608.06031`.

[2] Current workspace continuation, *A critical-margin upper bound with a remaining iterated logarithm*, `research/2026-10-27-minimax-audit/theorem/critical.tex`. Reviewed paper proof, not Lean-checked; preserved without modification.